

# MAURICIO MERCADO

## Senior Backend Engineer

Senior Backend Engineer specializing in scalable JavaScript/TypeScript systems, microservices architecture, and cloud infrastructure. 10+ years leading engineering teams, establishing technical standards, and integrating AI/LLM capabilities into production applications.

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## Skills & Competences

**Systems Architecture:** Design scalable, maintainable systems aligned with business goals; expert in microservices, modular monoliths, and event-driven architectures.

**Programming Languages:** TypeScript, JavaScript, Python, Go, Rust.

**Frontend Development:** React, Vue.js, Svelte, Expo (React Native); Tailwind CSS, React Query, Storybook — supporting experience used in consulting and staff-level roles.

**Backend Development:** Node.js (Fastify, Hono); proficient in building RESTful APIs and GraphQL for seamless system communication.

**Data Management:** PostgreSQL, MongoDB, Redis, SQLite, Elasticsearch; expertise in schema design, query optimization, and caching strategies.

**Cloud & Infrastructure:** AWS (Lambda, RDS, S3, EC2), Cloudflare, Docker, Pulumi/Terraform; experience with Kubernetes, serverless architectures, and multi-cloud deployments.

**BaaS & Modern Tools:** Git, Vite, CI/CD (GitHub Actions); automated testing (Vitest, Jest, Playwright).

**AI Integration:** LLM integration (OpenAI) for proof-of-concept features; AI-assisted workflow automation in experimental and consulting contexts.

**Technical Leadership:** Mentorship, code review excellence, establishing engineering standards, and guiding teams toward best practices with open-source collaboration principles.

## Work Experience

### Senior Software Engineer

#### Coursedog — 03/2023

- Stabilized 15+ university ERP integrations generating ~6 support tickets/week by improving field-level data handling, configurable retries, and dynamic header injection — cutting tickets to 1-2/week.
- Proposed and built @sisField/@sisEntity TypeScript decorator system, giving sales, marketing, and product teams live visibility into 600+ field mappings and directionality.
- Implemented OAuth2 integration architecture for LMS data retrieval (measurements, rubrics, submissions, assignments) with Vue.js frontend.
- Created video demos and comprehensive documentation for OAuth and LTI system integrations, improving team onboarding efficiency.
- Enabled non-technical staff to run local Coursedog instances for testing, reducing dependency on engineering resources.

### Founder And Principal Architect

#### Codigo — 01/2008

- Founded consulting practice delivering event-driven systems, API infrastructure, and automated provisioning using Node.js, TypeScript, AWS, React.js, and Svelte.
- Architected scalable systems from conception to deployment, making critical decisions on technology stack, infrastructure design, and scalability strategies.
- Designed cloud-native architectures leveraging AWS, Docker, and CI/CD, reducing deployment time by 75% and infrastructure costs by 30%.

### Staff Software Engineer

#### Sharecare (Previously Doc.Ai) — 03/2022 - 03/2023

- Redundant API calls made SmartOmix sluggish for field researchers on mobile — I replaced Redux with React Context + react-query caching, cutting API calls by 50% and accelerating feature delivery across 5 projects.
- Raised test coverage from 80% to 95%, reducing post-release bugs by 70% and giving the team confidence to ship faster.
- Built shared Storybook component library across 5 projects, eliminating duplicated code and enabling faster onboarding.
- Mentored team on advanced React patterns, TypeScript, and test-driven development approaches.

### Senior Engineer

#### Diligent (Previously Galvanize) — 10/2021 - 03/2022

- Test suite for 100,000+ daily-user app took ~45 min and crashed from memory exhaustion, blocking releases — I parallelized Jest and adopted Fastify + AWS Lambda, cutting run time from ~45 to ~18 minutes.
- Led Node.js testing standards across the team, preventing memory-heavy test patterns and enabling stable CI runs.
- Applied strangler pattern to modernize legacy sections incrementally, reducing regression risk.
- Mentored developers on JavaScript best practices and applied strangler patterns to modernize monolithic codebase.

### Lead Architect And Software Engineering Consultant

#### Mobilelive — 04/2021 - 10/2021

- Client's monolithic internal app was shared across 4+ teams who couldn't work independently — I architected a microfrontend solution using Webpack Module Federation, enabling each team to own and deploy their frontend separately.
- Built and open-sourced bolsterJS — a Webpack Module Federation abstraction enabling teams to share state and configure micro-frontends independently; proof of concept succeeded and client adopted the approach.
- Established deployment strategies using AWS CloudFront and defined inter-application communication patterns.
- Reduced time-to-market by 40% through modular architecture enabling parallel team development.

### Senior Lead Engineer

**Cto.Ai — 08/2020 - 04/2021**

- Built GitHub App integrating DORA metrics — the company's core product — giving leadership visibility into project health.
- Engineering teams had no visibility into container health — I deployed Go-based sidecar containers for real-time performance monitoring, reducing the sense of working blind that was driving employee churn.
- Drove open collaboration culture across teams, reducing project abandonment and improving cross-team velocity on AWS Serverless Framework infrastructure.
- Established automated quality control standards using GitHub Actions across 20+ projects.

### Senior Technology Architect

**Telus — 03/2019 - 07/2020**

- Teams were copying a stale Kubernetes template and couldn't stay current without complex rebases — I replaced it with a Fastify-based npm package so updates became as simple as running npm update, adopted by 15+ teams.
- Built cloud-agnostic deployment architecture spanning GCP and AWS, enabling consistent cross-cloud operations without vendor lock-in.
- Created multi-cloud deployment architecture supporting GCP and AWS with cloud-agnostic abstraction layer.
- Deployments previously took ~4 hours — I cut that to ~30 minutes through CI/CD workflow optimization and reduced onboarding time by 50% through comprehensive documentation.

### Senior Lead Engineer

**Navarik — 01/2019 - 03/2019**

- Team was migrating from a PHP monolith to Node.js event-driven architecture but was blocking the event loop, mixing async patterns, and building oversized React components — I led hands-on training for 18 developers to enable the migration without production regressions.
- Achieved 30% increase in code quality through improved review processes and testing infrastructure.
- Improved team productivity by 25% with 95% positive feedback on training relevance.

### Senior Engineer

**Humanapi — 07/2015 - 06/2018**

- Customers were churning and filing tickets because they couldn't find their clinics — I introduced Elasticsearch as a search layer replacing direct MongoDB queries, cutting search time from ~500ms to ~150ms and eliminating the support backlog.
- Implemented Elasticsearch search infrastructure, resulting in 70% improvement in search speed and accuracy.
- Rebuilt brittle test mocks with proper infrastructure, raising coverage from 65% to 92% and cutting release verification from 2+ hours to minutes.

### Principal Lead Engineer

**Selfie Inc. — 05/2013 - 07/2015**

- Users were churning saying "the app doesn't work" — timeline loads hit ~1s and when Wiz Khalifa joined, the spike made sluggishness highly visible; I redesigned around CQRS with async queues and fan-out caching using ZeroMQ and REST, cutting loads to ~200ms and scaling capacity 3x.
- Designed clear service boundaries using domain-driven design with circuit breakers, achieving 99.9% uptime for a platform handling 1M+ daily interactions.
- To handle burst traffic without data bottlenecks, I chose MongoDB for user schemas, Redis for caching (reducing DB load by 60%), and S3 for media storage.
- Built auto-scaling infrastructure on AWS reducing operational costs by 25% while supporting the 3x capacity increase needed to handle celebrity-driven traffic spikes.

## Education

**M.Sc. Computer & Software Engineering**

**Universidad De Los Andes, Colombia — 07/2011 - 03/2014**

**B.Sc. Electronics Engineer**

**Universidad Pontificia Bolivariana, Colombia — 06/2003 - 03/2007**